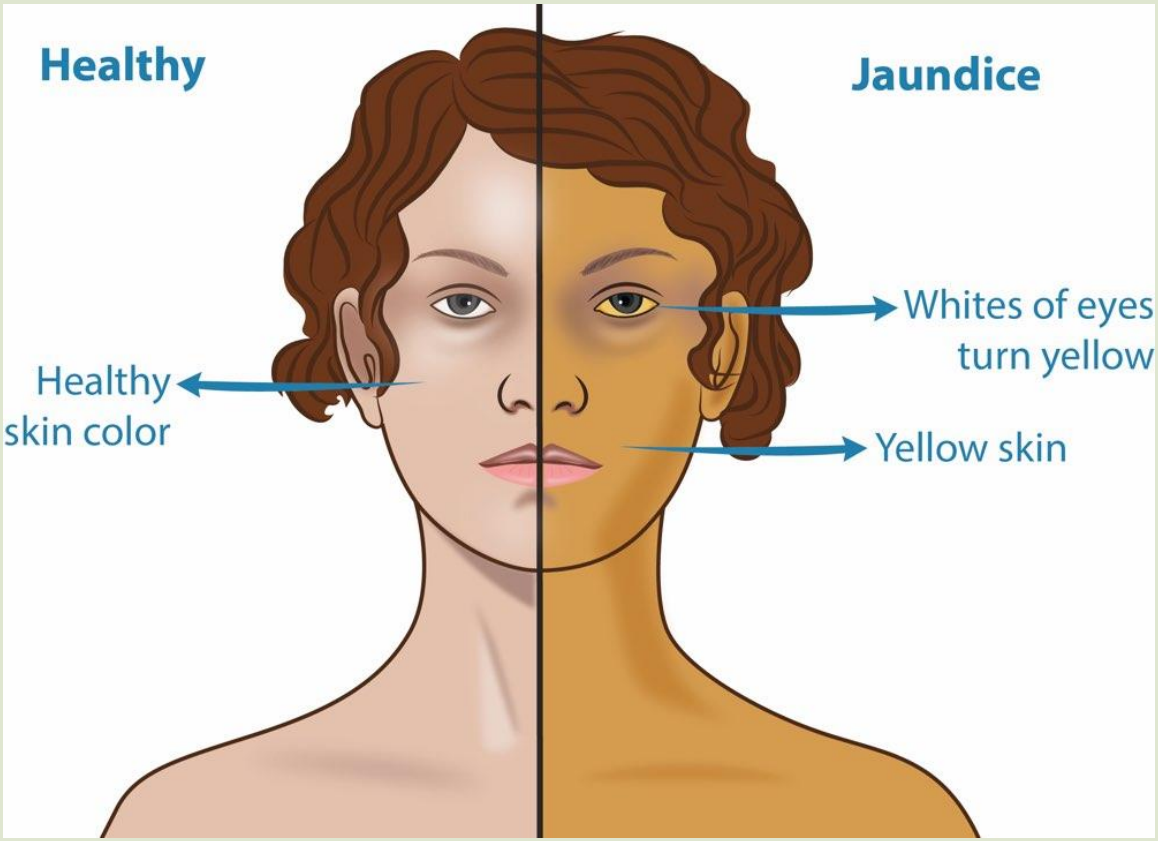
GIT
JAUNDICE

LEARNING OBJECTIVES

- Define jaundice
- Know normal levels of bilirubin
- Explain composition of bile
- Explain the normal physiology of bilirubin metabolism
- Identify Van den Bergh reaction
- Compare Types and causes of Jaundice

JAUNDICE

- **Yellowish discoloration of skin and mucus membranes; due to hyperbilirubinemia.**

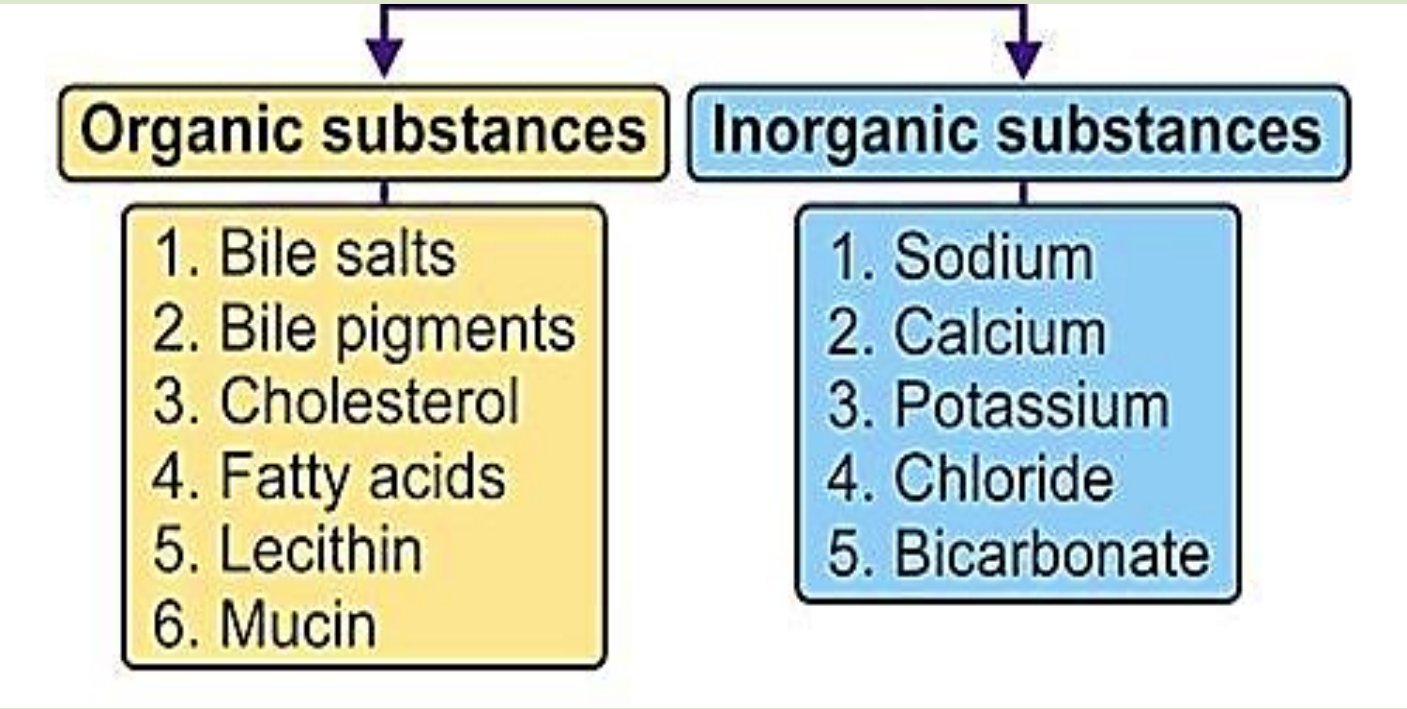


- First appear in sclera.
- Normal bilirubin = 0.5 to 1.5 mg/dl; yellowish color appear at > 2 mg/dl.

Bile

Composition of bile:

- 97% Water.
- 3% Solids.



BILE

* MADE in LIVER
* STORED in GALLBLADDER

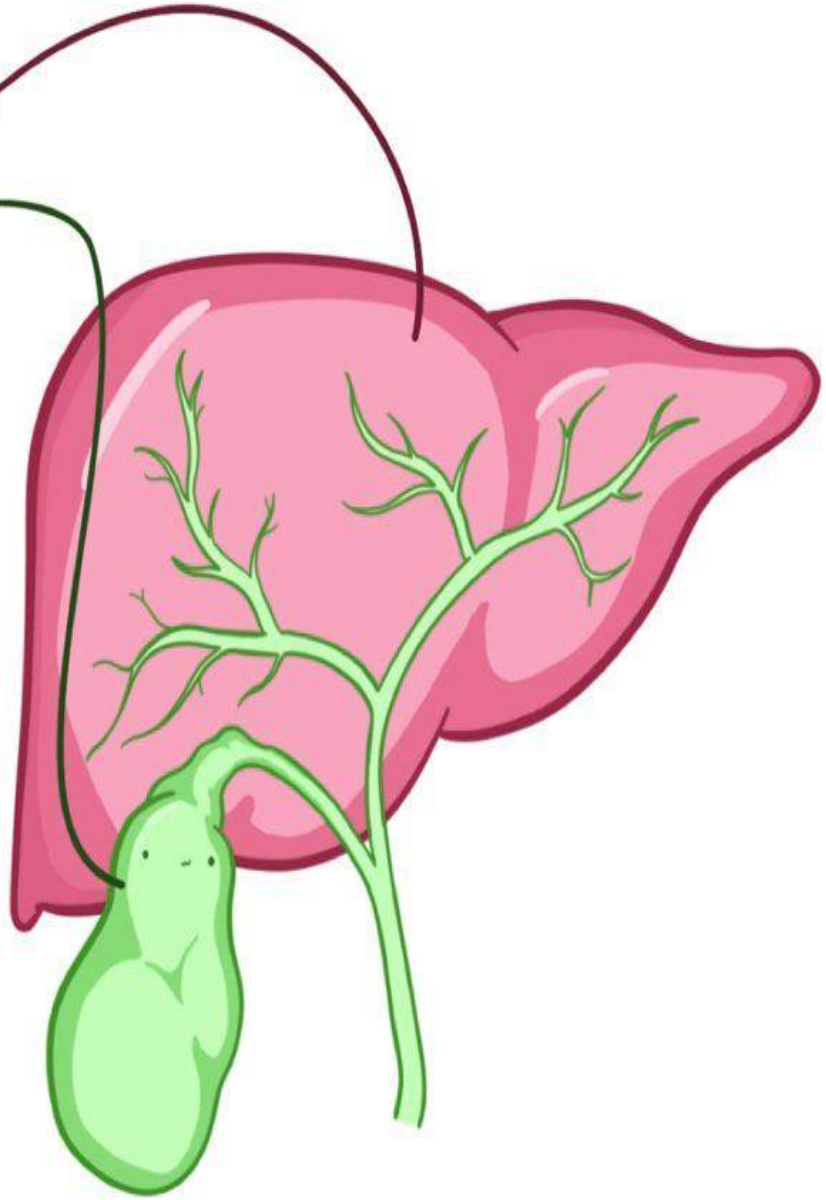
HELPS with...

* DIGESTION
* ABSORPTION of FATS (NORMALLY INSOLUBLE in WATER)
  ~ EMULSIFIES & SOLUBILIZES
* EXCRETION

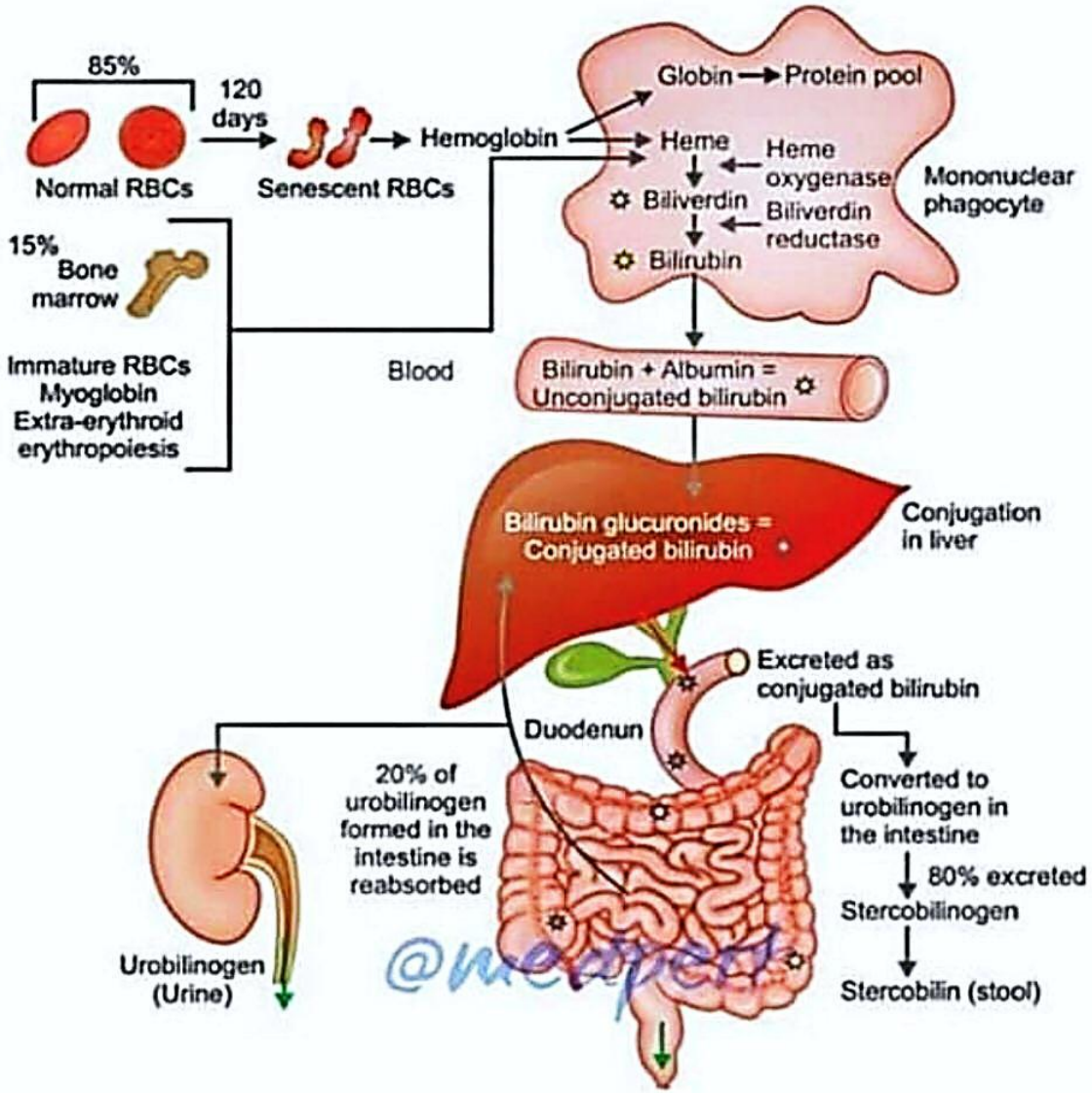
COMPOSITION

* BILE SALTS
* PHOSPHOLIPIDS
* CHOLESTEROL
* BILE PIGMENTS (BILIRUBIN)

(SMALL PERCENTAGE: CHOLESTEROL, BILE PIGMENTS)



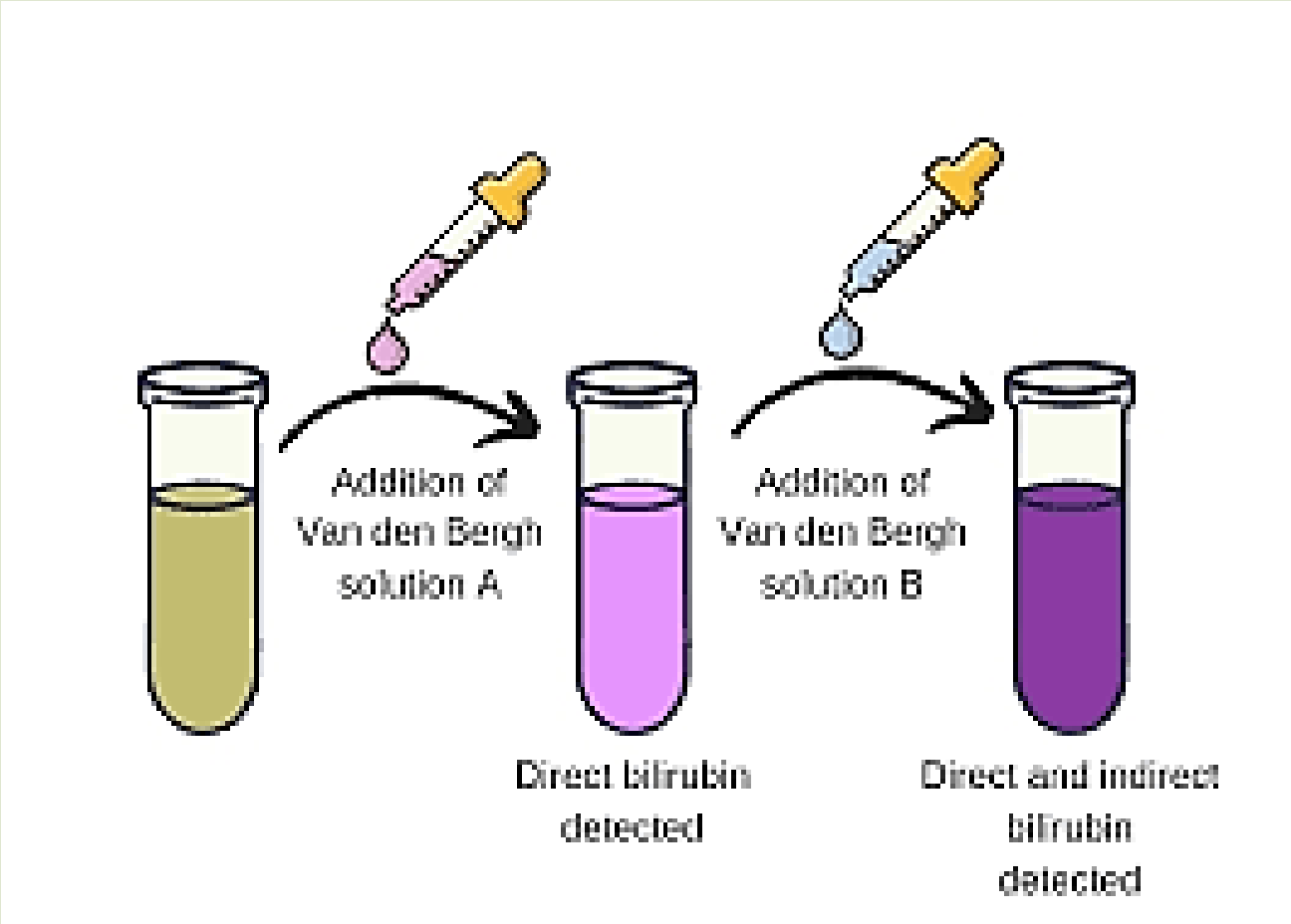
Bilirubin Metabolism



Van den Bergh reaction:

- Chemical reaction used to identify the type of jaundice.
- Principle: bilirubin reacts with diazo reagent to produce purple coloured azobilirubin.
- The serum of the patient is mixed with diazo reagent. If a red colour develops immediately it is called a direct bilirubin.
- In an indirect positive test, the patient's serum is first treated with alcohol and later mixed with diazo reagent. This causes development of a red colour. It is seen if unconjugated bilirubin is present.
- If both conjugated and unconjugated bilirubin are present the reaction is termed a biphasic reaction.

Van den bergh reaction



Types and causes of Jaundice

	Hemolytic	Obstructive	Hepatocellular
Causes	All causes of hemolysis: • **A**g-Ab react. • **B**acterial toxins + Malaria • **C**hemicals (Pb) • **D**rugs (Chemotherapy) • **E**nzyme **F**olate **G**6PD • Snake venom	• Stone • Cancer head of pancreas	• Viral hepatities • Cirrhosis
Type of bilirubin ↑	Haemobilirubin	Cholebilirubin	Both types
Van den bergh reaction	Indirect (Unconjugated)	Direct (Cojugated)	Biphase
Color of Pt.	Bale yellow	Very dark yellow	Yellow
Urine	Normal	Tea or cola-like	Tea or cola-like
Stool	Dark/Normal	Bale (with undigested fats)	Bale (with undigested fats)

A 43-year-old woman presents to the emergency department with the acute onset of abdominal pain. Her pain is located to the right upper quadrant (RUQ) and radiates to the right shoulder. She reports nausea and vomiting but no fever or chills. The RUQ pain is worse after she eats fatty meals. She reports a change in her urine and stool color.

On examination, the patient has severe RUQ tenderness. Her sclera is yellowish in color.

Investigations revealed that her white blood cell count is elevated, as are her liver function tests and alkaline phosphatase level. The amylase and lipase levels are normal. An abdominal ultrasound reveals an enlarged gallbladder with multiple stones and gallbladder wall thickening.

She subsequently is admitted to the hospital and undergoes a cholecystectomy.

Questions:

1- **What is The type of jaundice in this case?**
2- **What is the type of bilirubin that is mostly elevated in blood in this case?**
3- **What is the results of Van den Berg reaction?**
   a) **Compare between the types of jaundice regarding the color of urine,color of stool, and the skin color?**
4- **Which of the types of jaundice is the most associated with itching?**
5- **What is the cause of itching in such case?**

Thank you